

Soln 1: $A \rightarrow \text{Knight}$ $B \rightarrow \text{spy}$ $C \rightarrow \text{Knave}$

- Let A be the Knave. $\therefore$ C is not the Knave, since A will always lie.
- B says "A is the Knight" which is False. $\therefore$ B must be the spy, since a spy can lie.
- $\therefore$ C must be the Knight, but this contradicts with the statement "I am the spy" which is False. A Knight can never lie.

$\therefore$ A cannot be the Knave

- Let A be the spy. Case 1: Let C be the Knave. $\therefore$ B must be the Knight. B says "A is the Knight" which is False. It contradicts the fact that a Knight can never lie. $\therefore$ C cannot be the Knave.

Case 2: C must be the Knight. C says "I am the spy" which is False and a contradiction to the fact that a Knight can never lie.

$\therefore$ A cannot be the spy.

Final solution: $A \rightarrow \text{Knight}$ $B \rightarrow \text{spy}$ $C \rightarrow \text{Knave}$

29) A says "I am the Knight," B says "I am the Knave", and C says "B is the Knight."

Soln: A Knight can never lie. $\therefore$ He can never say "I am the Knave" because it is False. A Knave always lies. So, that person can never say "I am the Knave" because it is True. A spy, who can lie, can say "I am the Knave". $\therefore$ B must be the spy.

C says "B is the Knight", which is False. $\therefore$ C is the Knave, and hence A is the Knight.

Final solution: $A \rightarrow \text{Knight}$ $B \rightarrow \text{spy}$ $C \rightarrow \text{Knave}$